

# Keywords of Plotfunction

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## **1 Colors**

**1.0.1 beige**

**1.0.2 black**

**1.0.3 blue**

**1.0.4 brown**

**1.0.5 cyan**

**1.0.6 dblue**

Dark blue.

**1.0.7 dgreen**

Dark green

**1.0.8 gold**

**1.0.9 green**

**1.0.10 grey**

**1.0.11 lblue**

light blue

**1.0.12 lgreen**

light green

**1.0.13 lgrey**

light grey

**1.0.14 magenta**

**1.0.15 olive**

**1.0.16 orange**

**1.0.17 pink**

**1.0.18 purple**

**1.0.19 red**

**1.0.20 skyblue**

**1.0.21 violet**

**1.0.22 white**

**1.0.23 yellow**

## **2 Commands**

### **2.1 analyze:**

Usage: **analyze** [#nr = 1] [x<sub>1</sub> x<sub>2</sub>] [prec = 1e-9] [iterations = number of pixels of width][decimals = 4]

Analyses the function number nr in the [x<sub>1</sub> , x<sub>2</sub>] interval. prec is the precision used in the algorithms. Iteration is the number of evaluations of that function to count the changes of sign in the given interval. decimals are the number of decimals for rounding all results.

### **2.2 arclength:**

Usage: **arclength** [#nr = 1 / “function”] [decimals=4 iterations=50 precision=1e-8] [x<sub>1</sub> x<sub>2</sub>]

Calculates the arc length of the function #nr or “function” in the interval [x<sub>1</sub> x<sub>2</sub>] if given, or the x axis interval (default). The approximate result is rounded by the given decimals. Iterations and precision are needed in the used algorithms to approximate the integral value (Kepler Simpson).

### **2.3 binomtest:**

Usage: **binomtest** n; <= / = / >=; p; [ $\alpha$  = 0.05]; [color reject = red]; [color accept = green]

Computes and draws the binomial hypothesis test.

## 2.4 calculate:

Usage: **calculate arithmetic expression**

## 2.5 derive:

Usage **derive** [#nr = 1]

Derives function to x ( $x_1$ ).

## 2.6 evaluate:

Usage **evaluate** [#nr = 1] **expressions** [decimals = 4]

Evaluates the #nr-th. function value of the expressions and rounds the result to the given decimals.

## 2.7 integral:

Usage: **integral** [#nr = 1 / “term”] [decimals iterations precision] [x<sub>1</sub> x<sub>2</sub>] [v <color = lblue]

Numerical approximation (if possible also exact computation via a stamm-function) of the integral of the #nr-th function (or the function by the given term) in the interval [x<sub>1</sub> x<sub>2</sub>]. The result is rounded to the given decimals and the algorithms use the given number or iterations and precision (Kepler-Simpson-Algorithm).

With the option v enabled, the areas between the graph of the function and the x-axis in the given interval will be coloured in the given colour.

## 2.8 interpolation:

Usage: **interpolation** #nr / points for f [; points for f'] [;points for f'']

Polynomial interpolation of the #nr-th geometric object (points or polygon) or the given points for f and eventually f' and f''.

## 2.9 intersection:

Usage: **intersection** [#nr1 = 1] #nr2 [x<sub>1</sub> x<sub>2</sub>] [precision] [iterations] [decimals]

Computes the approximation (if possible also the exact) points of intersection and intersection angle between the #nr1-th and #nr2-th object in the [x<sub>1</sub> x<sub>2</sub>] interval. The coordinates are rounded to the given decimals. The Algorithm (Newton) uses the given precision and number of iterations.

### 2.10 latex-string:

Usage: **latex-string** [#nr = 1]

Translates the #nr-th function term to a latex string. Example: *latex-string*  $x/exp(x)$  would compute the string:  $\frac{x}{e^x}$

### 2.11 lower-sum:

Usage: **lower-sum** [#nr = 1]; n; a; b

Computes the lower sum of the function nr over [a,b] with n rectangles.

### 2.12 move:

Usage: **move** [#nr = 1] dx dy

Moves the #nr-th geometric object dx units to the right (left if  $dx < 0$ ) and dy units to the top (bottom if  $dy < 0$ ).

### 2.13 normal:

Usage: **normal** [#nr = 1] x / x y

Computes the normal to the #nr-th function at x or from the point (x,y).

### 2.14 osc-circle:

Usage: **osc-circle** [#nr = 1] x

Computes the osculating circle to the #nr-th function (or parametric curve) at x.

### 2.15 points-in-graph:

Usage: \*points-in-graph [#nr = 1] expression1 (x= / y=); expression2...  
[;dddecimals [= 4]] [T (with table)] \*

Draws the points in the graph of the nr-th function wrt the given x-, y- or parameter-expressions.

Example: *points-in-graph*  $x=4; y=sqrt(2); d=2$  draws the points to the graph of the first function wrt to  $x = 4$  and  $y = sqrt(2)$ . The number of decimals for these points is 2.

### 2.16 reflection:

Usage: **reflection** [#nr1 = 1] #nr2 / object

Reflects the object nr1 to the axis/point defined by the object nr2.

### 2.17 regression:

Usage: **regression** **#nr** [ = **1** ] / **points** [ **degree** = **regression-degree** ]  
Computes the regression polynomial of the given degree wrt the given points or the #nr-th geometric object.

### 2.18 reset-colors:

Resets all the colors to the default colors.

### 2.19 rotate:

Usage: **rotate** [ **#nr** = **1** ] **angle (in degrees)** **x y (center)**  
Rotates the #nr-th geometric object / function by an angle of the given degrees wrt the given center (x,y).

### 2.20 show-function:

Usage: **show-function** [ **#nr** = **1** ] **0/1**  
Shows (if value is 1) or hides (if value is 0) the #nr-th object.

### 2.21 show-grid:

Usage: **show-grid** **0/1**  
Shows (if value is 1) or hides (if value is 0) the grid.

### 2.22 show-x-axis:

Usage: **show-x-axis** **0/1**  
Shows (if value is 1) or hides (if value is 0) the x-axis.

### 2.23 show-x-scale:

Usage: **show-x-scale** **0/1**  
Shows (if value is 1) or hides (if value is 0) the x-scale.

### 2.24 show-y-axis:

Usage: **show-y-axis** **0/1**  
Shows (if value is 1) or hides (if value is 0) the y-axis.

### 2.25 show-y-scale:

Usage: **show-x-scale** 0/1

Shows (if value is 1) or hides (if value is 0) the y-scale.

### 2.26 stammfunction:

Usage: **stammfunction** [#nr = 1]

Tries to compute the stammfunction wrt ( $x_1$ ).

### 2.27 stretch:

Usage: **stretch** [#nr = 1] sx [x]/y ; [sy y]

stretches function nr by stretching factor sx among the x- and sy among the y-axis.

### 2.28 table:

Usage: **table** [#nr = 1 / "function"] [x<sub>1</sub> x<sub>2</sub>] [dx = 0.5] [;]

Evaluates the function in the values from x<sub>1</sub> to x<sub>2</sub> with an increment of dx.

If there is a ; at the end, the values are printed in fractions.

### 2.29 tangent:

Usage: **tangent** [#nr = 1] x / x y

Computes the tangent to the #nr-th function at x or from the point (x,y).

### 2.30 taylor-polynomial:

Usage: **taylor-polynomial** [#nr = 1] deg [x0 = 0]

Computes the taylor polynomial or deg-th degree of the nr-th function at x0.

### 2.31 upper-sum:

Usage: **upper-sum** [#nr = 1]; n; a; b

Computes the upper sum of the function nr over [a,b] with n rectangles.

### 2.32 zero-iteration:

Usage: \*zero-iteration [#nr = 1 / "function"] [decimals iterations precision]  
[x<sub>1</sub> x<sub>2</sub>] \*

Computes the approximation of a root in the  $[x_1, x_2]$  interval of the given object.

The result is rounded to the given decimals. The Algorithm (Newton) uses the given precision and number of iterations.

### 3 Constants

#### 3.0.1 inf:

stands for infinity.

#### 3.0.2 PI:

stands for the irrational number  $\pi$ .

## 4 Geometric Objects

### 4.1 bindensity:

Usage: **bindensity n p [style [transparency [width]]]**

Draws the binomial density wrt the variable  $X \sim \text{Bin}(n, p)$  ( $n > 0$  integer,  $0 < p < 1$ ) in different styles:

style = 0: points are drawn

style = 1: vertical lines are drawn

style = 2: filled rectangles are drawn with the given transparency (  $0 \leq \text{transparency} \leq 1$  )

where transparency = 0 is transparent and transparency = 1 is solid.

width is the width of the columns ( $\leq 1$ ).

### 4.2 bindistribution:

Usage: **bindistribution n p [style [transparency [width]]]**

Draws the binomial distribution wrt the variable  $X \sim \text{Bin}(n, p)$  ( $n > 0$  integer,  $0 < p < 1$ ) in different styles:

style = 0: points are drawn

style = 1: vertical lines are drawn

style = 2: filled rectangles are drawn with the given transparency (  $0 \leq \text{transparency} \leq 1$  )

where transparency = 0 is transparent and transparency = 1 is solid.

width is the width of the columns ( $\leq 1$ ).

### 4.3 bitmap:

Usage: **bitmap\_**[nr] x y width height

Draws the stored bitmap nr on to the position x y (axis-coordinates) with the given width and height in pixels.

### 4.4 circle:

Usage: **circle** x y r [ $\alpha_1$   $\alpha_2$  [slice]] [<color>]

Draws a circle with middle-point x y, radius r. If  $\alpha_1$ ,  $\alpha_2$  is given an arc of circle is drawn, beginning at the angle  $\alpha_1$  and ending at  $\alpha_2$ . If slice = 1 a transparent slice is drawn, if slice = 2 a filled slice is drawn.

### 4.5 fill:

Usage: **fill** x y [<color>]

Floodfill at the point (x,y) with the given color.

### 4.6 geodensity:

Usage: **geodensity** p [style [transparency [width]]]

Draws the geometric density wrt the variable  $X \sim \text{Geo}(p)$  ( $n > 0$  integer,  $0 < p < 1$ ) in different styles:

style = 0: points are drawn

style = 1: vertical lines are drawn

style = 2: filled rectangles are drawn with the given transparency ( $0 \leq \text{transparency} \leq 1$ )

where transparency = 0 is transparent and transparency = 1 is solid.

width is the width of the columns ( $\leq 1$ ).

### 4.7 histogram:

Usage: **histogram** points [t = 0 , w = 1 ] [<color>]

Draws a histogram wrt the given points. t is the value for the transparency of the background color, where 0 is transparent and 1 is opaque. w is the width of the columns wrt the x-axis range.

### 4.8 line

Usage: **line** point<sub>1</sub> point<sub>2</sub> [h = 0] [<color>] 1>

Draws a line (or arrow if h != 0) from point<sub>1</sub> to point<sub>2</sub>. h is a float number

for the size of the arrowhead ( $h = 0$  for no arrowhead). If  $l \geq 1$  the line will be dotted.

#### 4.9 **normaldensity:**

Usage: **normaldensity**  $\mu$   $\sigma^2$  [ **x1** ,**x2** ] [ <color>,<style> ]

Draws the normal density function wrt the variable  $X \sim \text{Norm}(\mu, \sigma^2)$ ,  
The rest of parameters are used as in usual functions.

#### 4.10 **normaldistribution:**

Usage: **normaldistribution**  $\mu$   $\sigma^2$  [ **x1** ,**x2** ] [ <color>,<style> ]

Draws the normal distribution function wrt the variable  $X \sim \text{Norm}(\mu, \sigma^2)$ ,  
The rest of parameters are used as in usual functions.

#### 4.11 **points:**

Usage: **points** **x1** **y1** **x2** **y2** ... **xn** **yn** [ <color> ]

Draws n points with the given coordinates.

#### 4.12 **poissondensity:**

Usage: **poissondensity** **lambda** [ <style> [ <transparency> [ <width> ] ] ]

Draws the Poisson density wrt the variable  $X \sim \text{Poi}(\text{lambda})$ ,  
where  $\text{lambda} > 0$ , in different styles:

style = 0: points are drawn

style = 1: vertical lines are drawn

style = 2: filled rectangles are drawn with the given transparency (  $0 \leq \text{transparency} \leq 1$  )

where transparency = 0 is transparent and transparency = 1 is solid.

width is the width of the columns ( $\leq 1$ ).

#### 4.13 **polygon:**

Usage: **polygon** **x1** **y1** **x2** **y2** ... **xn** **yn** [ <color> ]

Draws polygon with the n vertices  $(x_{1,y1}), \dots, (x_{n,yn})$ .

#### 4.14 **rectangle:**

Usage: **rectangle** **x1** **y1** **x2** **y2**

Draws a rectangle with the left bottom corner  $(x_{1,y1})$  and the top right corner  $(x_{2,y2})$ .

#### 4.15 **text:**

Usage: **text** {[l] string} x y [<color>]

Displays a string at the coordinates (x,y).

If latex is installed you can use `\to` generate a latex string.

#### 4.16 **triangle:**

Usage: **triangle** x<sub>1</sub> y<sub>1</sub> x<sub>2</sub> y<sub>2</sub> x<sub>3</sub> y<sub>3</sub> [<[color,] l>]

Draws a triangle with vertices (x<sub>1</sub>,y<sub>1</sub>), (x<sub>2</sub>,y<sub>2</sub>), (x<sub>3</sub>,y<sub>3</sub>).

If l >= 1 the line will be dotted.

## 5 Functions

### 5.1 **abs:**

Absolute value of a real or a complex number.

### 5.2 **arccos:**

Arcus cosine function.

### 5.3 **arcosh:**

Area hyperbolic cosine function.

### 5.4 **arcsin:**

Arcus sine function.

### 5.5 **arctan:**

Arcus tangent function.

### 5.6 **arsinh:**

Area hyperbolic sine function.

### 5.7 **artanh:**

Area hyperbolic tangent function.

### **5.8 cos:**

### **5.9 cosh:**

Hyperbolic cosine.

### **5.10 exp:**

The exponential function.

### **5.11 log:**

Logarithmus Naturalis.

### **5.12 sin:**

Sine function.

### **5.13 sinh:**

Hyperbolic cosine.

### **5.14 sqrt:**

The square root function.

### **5.15 tan:**

The trigonometric tangent function.

### **5.16 tanh:**

Hyperbolic tangent function.

## **6 Settings**

### **6.1 axis-color:**

Usage: **axis-color** / **Red Green Blue**

Use with the defined colors or input three values from 0 to 255 for the red, green and blue part.

## 6.2 axis-fontsize:

Usage **axis-fontsize** unsigned int

## 6.3 axis-names:

Usage: **axis-names** string string

Example: *axis-names x y*

## 6.4 axis-range:

Usage: **axis-range**  $x_1$   $x_2$  [:  $y_1$   $y_2$ ]

Example: *axis-range -3 5 : -2 7* : x values range from -3 to 5, y values from -2 to 7  
*axis-range 5 : 10* : x values from -5 to 5, y values from -10 to 10.

## 6.5 axis-scale:

Usage : **axis-scale**  $s_x$  [ $s_y = s_x$ ]

Example: *axis-scale  $\pi/2$  1*

## 6.6 axis-width:

Usage: **axis-width** unsigned int

thickness of axis lines in pixels

## 6.7 background-color:

Usage: **background-color** def. color / Red Green Blue

Use with the defined colors or input three values from 0 to 255 for the red, green and blue part.

## 6.8 default-color:

Usage: **default-color** def. color / Red Green Blue

Sets the default color for objects when multicolored is off.

## 6.9 font-size:

Usage: **font-size** unsigned int

Sets the font size in pixels for input/output text.

### 6.10 function-color:

Usage: **function-color** [#nr = 1] **defined color** / **Red Green Blue**

Draws the #nr-th function/geometric object with the given defined color or the color defined by the three values from 0 to 255 for the red, green and blue part.

### 6.11 function-settings:

Usage: **function-settings** [#nr = 1]

Opens a settings dialog for the #nr-th. function or geometric object.

### 6.12 function-size:

Usage: **function-size** [#nr = 1] **unsigned number**

Sets the line thickness of the #nr-th. function/geometric object to the given number.

### 6.13 grid-color:

Usage: **grid.-color** **def. color** / **Red Green Blue**

Draws the grid lines in the given defined or the color obtained by the three values for the red, green and blue part (from 0 to 255 each).

### 6.14 grid-scale:

Usage: **grid-scale** **sx** [**sy = sx**]

Example: *grid-scale*  $PI/2$  1

### 6.15 grid-width:

Usage: **grid-width** **unsigned int**

thickness of grid lines in pixels

### 6.16 margin:

Usage: **margin** **unsigned int**

Sets the margin (offset) of the coordinates system in pixels.

### **6.17 move-increment:**

Usage: **move-increment** dx [dy=dx] (p for pixels)

Move increment values for moving objects or the coordinate system with the arrow-keys.

### **6.18 multicolored:**

Usage: **multicolored** 1/0

Toggle multicolored objects.

### **6.19 panel-size:**

Usage: **panel-size** sx [sy = sx]

### **6.20 parameter-values:**

Usage: **parameter-values** double ...

Sets the values for the parameter t of the functions.

### **6.21 point-decimals:**

Usage: **point-decimals** int

### **6.22 point-size:**

Usage: **point-size** int

Sets the size in pixel for points (wrt to style 1 and 2)

### **6.23 regression-degree:**

Usage: **regression-degree** unsigned int

Sets the regression degree for the regression computations.

### **6.24 select:**

Usage: **select** [#nr=1]

Selects (highlights) the nr-th function.

### **6.25 values-number:**

Usage **values-number unsigned int**

The number of points that are evaluated for rendering the functions.



## 7 Alphabetically

### 7.1 A

7.1.1 abs

7.1.2 analyze:

7.1.3 arccos

7.1.4 arclength

7.1.5 arcosh

7.1.6 arcsin

7.1.7 arctan

7.1.8 arsinh

7.1.9 artanh

7.1.10 axis-color

7.1.11 axis-fontsize

7.1.12 axis-names

7.1.13 axis-range

7.1.14 axis-scale

7.1.15 axis-width

### 7.2 B

7.2.1 background-color

7.2.2 beige

7.2.3 bindensity

7.2.4 bindistribution

7.2.5 binomtest

7.2.6 bitmap

7.2.7 black

7.2.8 blue

7.2.9 brown

### 7.3 C

7.3.1 calculate

7.3.2 cos

7.3.3 cosh

7.3.4 cyan